COMP90042 NLP

L23: Subject Review (Part 1)

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Preprocessing

- Sentence segmentation
- Tokenisation
  - Subword tokenisation
- Word normalisation
  - Derivational vs. inflectional morphology
  - Lemmatisation vs. stemming
- Stop words

N-gram Language Models

- Derivation
- Smoothing techniques
  - Add-*k*
  - Absolute discounting
  - Katz Backoff
  - Kneser-Ney smoothing
  - Interpolation

Text Classification

- Building a classification system
- Text classification tasks
  - Topic classification
  - Sentiment analysis
  - Native language identification
- Algorithms
  - Naive-Bayes, logistic regression, SVM
  - kNN, neural networks
- Bias vs. variance

Part-of-Speech Tagging

- English POS
  - Open vs. closed POS classes
- Tagsets
  - Penn Treebank tags
- Automatic taggers
  - Rule-based
  - Statistical
    - Unigram, classifier-based, HMM

Hidden Markov Models

- Probabilistic formulation
  - Parameters: emission and transition probabilities
- Training
- Viterbi algorithm
- Generative vs. discriminative models

Feed-forward Networks

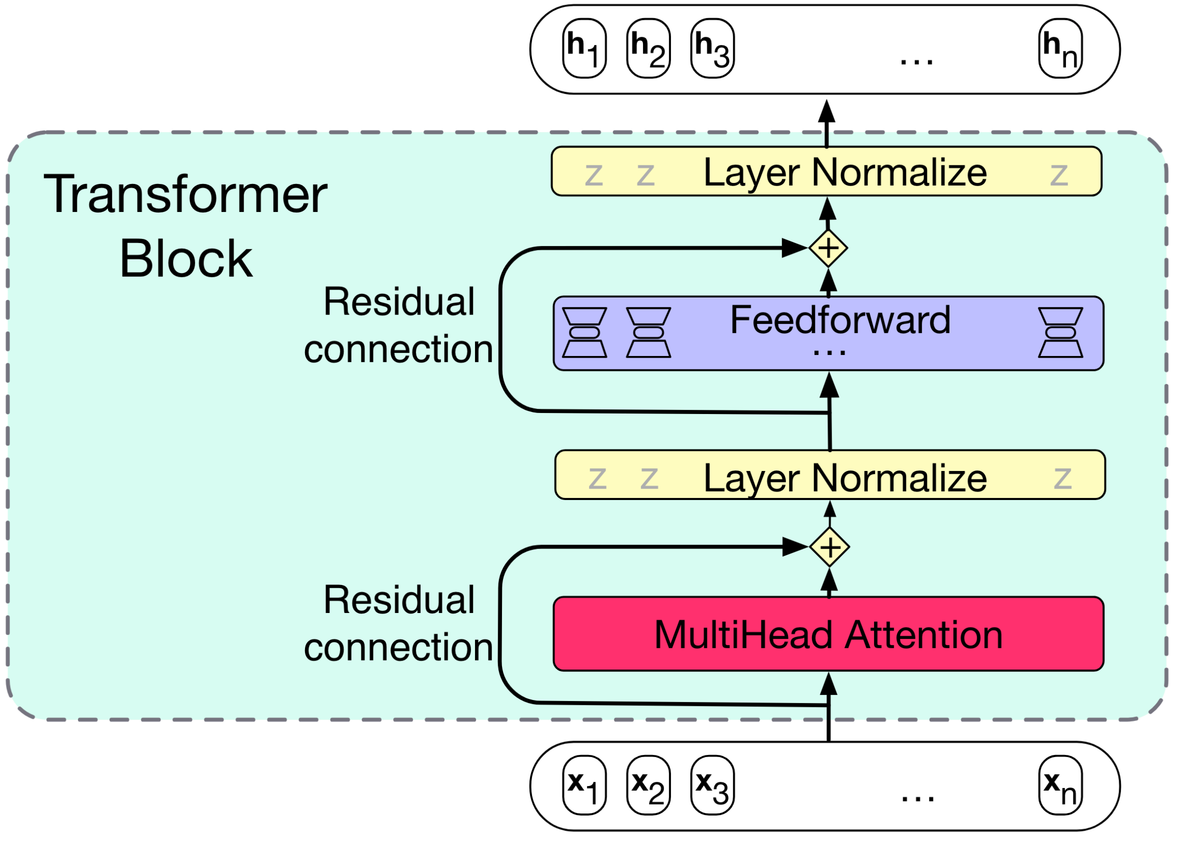
- Formulation
- Designing FF networks for NLP tasks
  - Topic classification
  - Language model
  - POS tagging
- Word embeddings

Recurrent Networks

- Formulation
- RNN language models
- LSTM
  - Functions of gates
  - Variants
- Designing RNN for NLP tasks
  - Text classification: sentiment analysis
  - POS tagging

Transformer

- Formulation with pure attention
  - Query/Key/Value Comparison
- Transformer block
- Position embeddings



Word Embeddings

- Matrices for distributional semantics
  - VSM, TF-IDF, word-word co-occurrence
- Association measures: PMI, PPMI
- Count-based methods: SVD
- Neural methods: skip-gram, CBOW
- Evaluation
  - Word similarity, analogy

Pretrained Models

- GPT
  - Decoder model; pros/cons
- BERT
  - Encoder model; masked language model; pros/cons
- T5
  - Encoder-decoder model; training objectives: span corruption

Exam

Exam Structure

- On-campus, closed book
- Non-programmable calculator is permitted
- 120 points (40% for the subject)
- Time: 120 minutes writing time + 15 minutes reading time
- 3 parts:
  - A: short answer questions
  - B: method questions
  - C: algorithm questions

Short Answer Questions

- Several short questions
  - 2-3 sentence answers for each
  - Definitional, e.g. what is X?
  - Conceptual, e.g. relate X and Y, purpose of Z?
  - May call for an example illustrating a technique/ problem

Method Questions

- Longer answer
- Focus on analysis and understanding
  - Contrast different methods
  - Analyse an algorithm/application
  - Motivate a modelling technique
  - Explain or derive mathematical equation

Algorithmic Questions

- Perform algorithmic computations
  - Numerical computations for algorithm on some given example data
  - Present an outline of an algorithm on your own example

What to Expect

- Even coverage of topic from the semester
- Be prepared for concepts that have not yet been assessed by assignments / project
- Sample exam on LMS (Modules > Sample Exam)